

Ryan **Slocum**

MSc Robotics, Systems and Controls · ETH Zurich
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Education

ETH Zurich — *Zurich, Switzerland*
MSc ETH in Robotics, Systems and Controls — *09/2024 – 03/2027 (expected)*
- Coursework: Perception and Learning for Robotics, Probabilistic AI, Machine Perception, Planning and Decision Making, Recursive Estimation

University of Colorado Boulder — *Boulder, CO, USA*
BSc in Computer Science, GPA 3.99/4.0, minor in Leadership Studies — *08/2020 – 05/2024*

Experience

ANYbotics — *Zurich, Switzerland*
Perception Software Engineer Intern — *02/2026 – 07/2026*
- Built an automated SLAM evaluation pipeline for the ANYmal legged robot by replaying a ground-truthed benchmark via GitLab CI
- Migrated 15+ SLAM and mapping packages from ROS1 to ROS2 in production, written in C++ and Python
- Installed and integrated an OptiTrack motion-capture system to record new ground-truth scenarios

Outrider — *Brighton, CO, USA*
Software Engineer Intern, Test — *05/2023 – 07/2023*
- Led migration of safety-firmware tests to Python on a Hardware-in-the-Loop rig validating e-stop and steering controllers

Company Six — *Boulder, CO, USA*
Software Engineer Intern — *05/2022 – 02/2023*
- Owned the iOS teleoperation app for first-responder robots, taking it from proof of concept to near-release with an intuitive steering scheme, multi-robot monitoring, and push-to-talk
- Wrote low-level robot behaviors in embedded C, e.g. an autonomous surveillance mode

Research and Projects

MSc Thesis, ETH Robotic Systems Lab — *Zurich, Switzerland*
Reinforcement Learning, Legged Locomotion — *09/2026 – 03/2027 (expected)*
- Integrating a model of thermal dynamics for quadruped robots into the reinforcement-learning pipeline to learn to avoid overheating and self-damage

Semester Project, ETH IDEAL Lab — *Zurich, Switzerland*
LLMs, Agentic AI — *09/2025 – 12/2025*
- Explored approaches for using generative AI to create robot-assemblable furniture designs from text, including fine-tuning with SFT and GRPO

Bird's-Eye-View Mapping, ETH Robotic Systems Lab — *Zurich, Switzerland*
PyTorch, Deep Learning — *02/2025 – 06/2025*
- Trained a model to predict semantic and elevation maps from multi-camera images for mobile robotics; trained over ∼1 TB of data on HPC cluster

BSc Thesis, CU Boulder ARPG — *Boulder, CO, USA*
Visual SLAM, C++ — *08/2023 – 05/2024*
- Extended Direct Sparse Odometry with an onboard-illumination model to reduce tracking error for visual SLAM in dark environments

Pathly, ETH Entrepreneurs Club Hackathon — *Zurich, Switzerland*
1st Prize, CHF 5,000 — *11/2025*
- Built and pitched an AI-agent outdoor trip planner with a live demo in 24 hours, team of four

Skills

Programming	C++, Python, C
Robotics & ML	ROS/ROS2, SLAM, Reinforcement Learning, PyTorch
Tooling	Git, GitLab CI, Docker, HPC
Languages	English (native), German (B2)